

# Development of a new nano modified cement based adhesive for FRP strengthened RC members

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## HIGHLIGHTS

- Mechanical properties and bond strength of CBA are improved using nano material.
- GNP, GO and NS refine the pore structure of the CBA and reduce the chloride depth.
- GNP mix achieved full utilisation of the FRP textile ruptured at ultimate capacity.
- The use of FRP textile resulted in higher bond properties for all CBA mixes.

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## ABSTRACT

There is currently a high demand in the construction industry for high-strength and durable cement-based adhesives (CBAs) for use in a variety of applications such as ground anchors, ground improvement, concrete repair or as a substitute for epoxy resins to bond carbon fiber reinforced polymers (CFRPs) to concrete structures where increased fire performance is required. Limited research has been conducted on the use of nano-materials such as graphene oxide (GO), graphene nanoplatelets (GNPs) and nano silica (NS) to improve the mechanical properties and durability of CBA. This paper presents a new mix design comprising ordinary portland cement, GNP, GO and NS to improve the strength and durability of CBA. Furthermore, the paper presents bond-slip measurements between carbon fibre reinforced polymer (CFRP) and CBA embedded in concrete prisms. Single lap shear prisms are tested using different CBA mixes and structural epoxy to assess the bond for near surface-mounted (NSM) strengthened structures. The results show improvement in the compression and flexural strength of the CBA by up to 46% and 57% respectively by using NS and GO at low concentrations. The durability tests indicate that the modified CBA has 61% lower chloride penetration than standard CBA due to reduced porosity. The pull-out tests were conducted of FRP-to-concrete joints using the CBA materials and results of GNP-reinforced CBA show an improvement of 27% and 73% for CFRP laminate and textile, respectively.

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## 1. Introduction

Carbon fiber reinforced polymers (CFRP) have several well-documented advantages over traditional strengthening materials, including its light weight, high tensile strength, durability, ease of installation and unobtrusiveness. The application of CFRP has been found to be more cost-effective than traditional strengthening methods and may negate the need to replace the original structure. The cost-effectiveness of the application of CFRP composites is further increased by the ongoing savings associated with the lack

of maintenance. As a result, CFRP strengthening enjoys a great deal of popularity in the construction industry. The use of CFRP reinforcement is having a significant impact on the economics of bridge maintenance and rehabilitation worldwide, and in many cases permits an increase in the load-carrying capacity of bridge members as opposed to replacement. A recent example is Melbourne's Westgate Bridge, which was built to support 40,000 vehicles per day and was recently strengthened in order to accommodate 140,000 vehicles per day [1]. CFRP has also been extensively implemented to strengthen reinforced concrete (RC) structural elements in buildings, the majority of which require strengthening due to construction or design defects or changes in use resulting in increasing loading, as was the case for the White City shopping centre in Shepherd's Bush, London [2].

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However, the current widespread use of epoxy and organic resins to bond FRP materials to concrete members has significant limitations in high-temperature environments, due to the low glass transition (TG) temperatures, which are in the range of 60–70 °C. When temperatures exceed this range, the strength and stiffness of organic adhesives degrade rapidly and they no longer retain their performance characteristics. This poses significant problems in certain parts of the world such as Saudi Arabia, where ambient temperatures can reach up to 52 °C and surface temperatures can be much hotter when exposed to direct sunlight. Further, all structures are required to maintain a minimum degree of structural performance in the case of a fire to ensure the safety of occupants and allow evacuation of the structure without collapse. Unfortunately, the use of epoxy resins to bond CFRP to structural elements and their low TG properties mean that the composite action and strength contribution that CFRP adds to the structure is lost in a fire. This can pose significant problems for buildings and is difficult to resolve, even with the use of thermal insulation and fire protection systems. Organic resins also have problems relating to flammability, emission of toxic fumes and steroids during curing and moisture permeability, which make them less than ideal as bonding agents. In order to address the limitations of epoxy based systems in CFRP applications, researchers have explored both cement based adhesives and flexible polyurethanes. Fiber reinforced polyurethanes (FRPU) were found to address the brittle nature of epoxy resins and increase the ductility of the strengthened structure [3,4]. Polyurethane materials also have higher glass transition temperatures of approximately 165 °C compared to 60 °C for epoxy resins which may reduce the amount of insulation to the FRP required in a fire load case.

Cement-based adhesives (CBAs) may be suitable alternative epoxy resins for bonding FRP to concrete members due to their inherent properties, which include no emission of toxic fumes during mixing or curing, non-flammability and excellent performance when subjected to high temperatures such as fire [5]. However, it is well known that cement-based materials are quasi-brittle in nature, have low tensile strengths, are prone to cracking, and possess low toughness. Compared with organic adhesives, the use of cement-based materials as bonding agents increases the likelihood of premature debonding failure, either through cohesive failure within the cement-based adhesive itself or inter-laminar failure at the interface between the FRP-adhesive or the adhesive-concrete due to the weak nature of the bond. However, the introduction of nano-materials (NMs) in CBAs such as graphene oxide (GO), graphene nano-platelets (GNPs), and nano-silica (NS) has the potential to significantly enhance the mechanical properties and durability of CBAs and potentially replace the use of organic resins in FRP strengthening applications.

Research has been underway over the past decade to improve the mechanical performance and durability of CBAs. This work has involved the introduction of nano-materials (NMs) in the cementitious paste in order to bridge the formation of micro-level cracking, thus improving its tensile strength properties. Other applications involve improving the durability of CBA by the use of NM to fill pores within the cementitious matrix, which reduces the porosity of the material and provides better protection of steel reinforcement against the ingress of moisture and corrosion. Many NMs have now been developed and introduced into cementitious materials, including graphene oxide (GO), graphene nano-platelets (GNPs), nano-silica (NS), carbon nanotubes, and nano-alumina [6–9].

Generally, NMs are classified into three categories depending on their size and morphology: (1) zero-dimensional (0-D) NMs have very low aspect ratios, which have proved to be ineffective in arresting micro-level cracks within the cementitious matrix; (2) one-dimensional (1-D) NMs such as carbon nanotubes may effec-

tively arrest the formation of micro-cracking but tend to agglomerate in the cement matrix, making uniform dispersion a challenge [10]; (3) two-dimensional (2-D) NMs such as graphene and graphene oxide possess excellent mechanical properties due to their sp<sup>2</sup>-bonded carbon atoms [11]. Further, it has been reported that graphene oxide can be well dispersed in the cementitious matrix, making it the most promising NM for improving the mechanical performance and durability of the cementitious matrix [12]. Graphene-enhanced CBA possesses several advantages over traditional cementitious materials, including (1) higher tensile and compressive strengths; (2) lower permeability resulting from smaller pore sizes; (3) its high early strength accelerates construction times; (4) improved electromagnetic shielding and (5) improved fire resistance due to increased thermal conductivity. However, further research is required to optimise the properties of graphene-enhanced CBA and extend its usefulness into other civil engineering fields, such as its use as a CBA for bonding CFRP to concrete members for strengthening applications.

Currently, the mechanical properties of graphene-modified CBA are not sufficiently refined to enable the substitution of organic resins for externally-bonded FRP applications. Pan et al. [12] reported that 0.05% GO increases flexural strength by up to 41–58%. Further, Lu et al. [13] concluded that the addition of 0.08 wt% GO leads to a 24.8% increase in compressive strength and a 37.7% increase in tensile strength while ultimate strain remains unchanged, an 80.6% increase in flexural strength and a 105% increase in flexural toughness. These findings were confirmed by Zhao et al. [14], who reported the similar findings that the addition of 0.022 wt% GO combined with 0.22 wt% cement increased the compressive strength, flexural strength, Young's modulus and flexural toughness by 34.10%, 30.37%, 32.37% and 33%, respectively. Lu et al. [13] also pointed out that the functional groups (–COOH or –OH) of GO increase the bond with calcium-silicate-hydrates.

The above results offer reasonable enhancements to the mechanical performance of cement mortar. On the other hand, NS is considered to be more stable in a water solution than graphene products. Colloidal silica can be added directly to the mix to improve the mechanical properties and durability of high-performance concrete [15]. NS acts as a nano-filler and contributes to the pozzolanic reaction. As a result, pore structures are refined and the concrete's durability is improved [16]. NS in concentrations of up to 3 wt% has also been found to be beneficial in improving the thermal resistance of cement mortars when subjected to elevated temperatures [17]. Horszczaruk et al. [17] attributed this to NS improving the cement matrix by the formation of additional C–S–H bonds which prevented crack propagation during high temperature exposures.

Many factors affect the properties of cement composites strengthened by nano-materials. Surface area (or particle size) is considered to be an important factor in improving the properties of cement composites [18]. The mixing and dispersion methods of graphene products have different effects on cement composites. Most researchers recommend that graphene products should be dispersed in water homogeneously by using sonication and surfactants. An et al. [19] suggest that the addition of 0.05% of GO and sonication with water produce considerable improvements in the compressive and flexural strengths of the concrete at 28 days by 25.5% and 37.3%, respectively. However, mixing GO with cement in powder form has an effect on improving the mechanical properties of the mix, as the compressive strength and flexural strength of the concrete at 28 days increase by 16% and 27%, respectively [19].

ISO 527-5 [20] stipulates that the minimum tensile strength of adhesive which should be accepted for externally bonded FRP applications is 12 MPa. However, recent research has found that the tensile strength of the adhesive is not the only parameter

resulting in efficacy in bonding FRP to concrete. Sometimes lower strength but highly deformable and durable adhesives are more effective [21,22]. Further, the tensile strength of the adhesive is less critical in NSM FRP applications which have a higher bond surface area between the adhesive and the substrate and FRP. In these cases, the tensile strength of the substrate is usually more critical.

However, all direct tensile strength results reported to date for graphene-enhanced mortars fall short of this value, although flexural tensile strengths of up to 11.1–14.2 MPa have been reported for GO CBA [13,14]. Another key parameter which requires optimisation for the bonding of CFRP to concrete is the bond strength of the CBA. Mohammed et al. [23] conducted pull-off tests to assess the bond behaviour of graphene-modified cements compared with general cement mixes. The results of this study indicated that graphene-modified cements showed a 78% improvement in bond strength compared with normal cement mixes. The results also indicated that the graphene mix exhibited a pull-off strength of approximately 74% of the pull-off strength of samples bonded using epoxy resins. Although these results are based on limited studies, there is evidence that the development of a graphene-modified cement with sufficient tensile and bond strength for CFRP applications is a feasible endeavour which requires further research.

Therefore, the aims of the project are as follows:

- To develop a new cement-based composite adhesive (CBA) to be used as a bonding agent with the use of nano-additives such as graphene oxide (GO), graphene nanoplatelets (GNPs) and nano-silica (NS). The CBA will be formulated for superior tensile strength, bond characteristics, fire performance and durability. The enhancement mechanism of the nano-additives will be characterised based on experimental testing.
- To evaluate the performance of CBA when used to bond fibre-reinforced polymer composites (FRPs) to concrete using near-surface mounted (NSM) techniques based on joint level tests. These tests will be used to quantify the bond-slip properties between the CBA, the FRP and the concrete interfaces.

## 2. Research significance

The research significance is to develop a new nano modified cement based adhesive (CBA) with superior tensile strength, bond strength, durability and fire resistance which can be used to bond FRP to concrete in near surface mounted (NSM) applications. The new CBA material will address the limitations of low glass transition temperatures and poor fire performance of epoxy resins and providing a strengthening solution which is cost effective, efficient and durable.

### 2.1. Experimental work

To meet the requirements for adhesive, modifications need to be made to cement-based composites, such as lowering the water-to-cement ratio or adding fine particles to densify the microstructures. Recently, with the rapid development of technology, graphene and graphene-based products have been demonstrated to effectively improve the mechanical performance of cement-based composites [19,24–26]. The reinforcing mechanism can be attributed to the interaction between the nano-particles and the cement hydration products, as well as the effect of nucleation on cement hydration acceleration. However, even with the advantages of these nano-particles, the tensile strength is still lower than required. The hybrid use of nano-silica (NS) and graphene oxide (GO) is explored in this section to determine the influence of 0-D and 2-D nano-particles on the compressive and tensile strengths of cement-based composites. Based on previous research,

up to 2% NS and 0.5% GO by weight of cement was added to cement-based composites, either individually or collectively, to determine their influence on microstructure, strength and durability. To ensure the uniform dispersion of NS and GO, a combination of ultra-sonication and chemical surfactant was employed prior to cement mixing. A previous study determined that polycarboxylate-based super-plasticizer is a very effective surfactant to disperse graphene products [27]. The amount of super-plasticizer was selected to maintain the same flowability and the time of homogenization was minimized.

#### 2.1.1. Materials

The materials used in this study were general-purpose ordinary portland cement (OPC), graphene oxide (GO), graphene nanoplatelets (GNPs) and nano-silica (NS). The following section outlines the properties of each of these materials.

General purpose cement sourced from Cement Australia was used in all mixes. The cement complied with the requirements of AS 3972 [28] and had a mean particle diameter of 12.96  $\mu\text{m}$ . The particle size distribution is presented in Fig. 1.

Graphene oxide (GO) was supplied by Garmor Tech in the form of a dry powder with a nominal particle size of 500 nm, and 90% of the GO particles were below 800 nm in size. Moreover, more than 90% of the particles had an average thickness of 10 layers of graphene. The GO powder was composed of 99.8% pure carbon.

The GNPs used in this study were supplied by Grapheneca in the form of an aqueous graphene paste. As shown in Table 1, the aqueous graphene paste consisted of 32% pure GNP and 68% water which had been premixed. Sonication of the paste was trialled using 6 samples, each sonicated for 2 min at 1000 W using a probe sonicator. The samples were compared with unsonicated specimens in terms of compressive strength and tensile strength and sonication in this manner was found to not influence the results. As a result, the graphene paste was mixed with water to achieve the desired water-cement ratio and then added to the mix without sonication.

The nano-silica (NS) was supplied by BASF in the form of a commercial colloidal silica called MasterRoc 695. This product contains 14% by weight of NS with particle sizes ranging from 15 to 55 nm. In order to increase the workability of mixtures containing a higher percentage of NS, a polycarboxylic superplasticiser (MasterGlenium SKY 8100) was used.

The FRP laminates used in the FRP-to-concrete bond tests presented in a later part of the paper were 1.4 mm thick  $\times$  20 mm

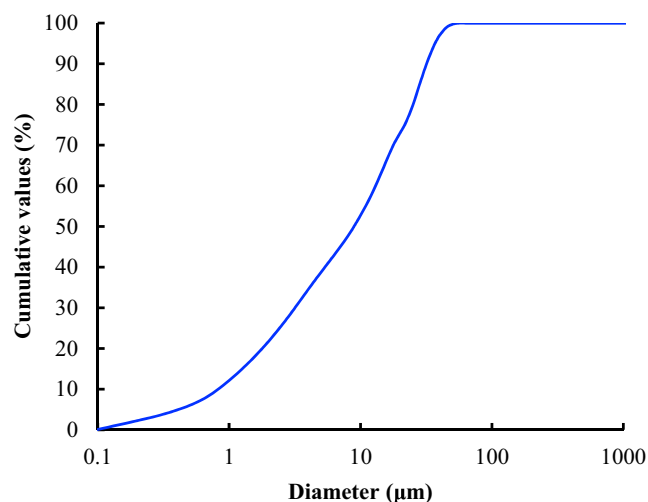


Fig. 1. Particle size distribution of OPC cement.

**Table 1**  
Graphene Paste Composition.

| Water<br>% | Graphene Nano-platelets 32%     |                   |                                           |                                       |               |               |                   |
|------------|---------------------------------|-------------------|-------------------------------------------|---------------------------------------|---------------|---------------|-------------------|
|            | Flake Size<br>( $\mu\text{m}$ ) | Thickness<br>(nm) | Surface Area<br>( $\text{m}^2/\text{g}$ ) | Density<br>( $\text{g}/\text{cm}^3$ ) | Carbon<br>(%) | Oxygen<br>(%) | Impurities<br>(%) |
| 68         | 2–6                             | <3                | 536                                       | 0.02                                  | >99.4         | <0.5          | <0.1              |

wide. To assess the mechanical properties of the FRP laminates, the tensile strength and modulus of elasticity were determined in accordance with ASTM D3039 [29]. A total of 5 laminate coupon specimens were prepared and tested and the results indicated an average tensile strength of 3300 MPa with a standard deviation of 79.3 MPa. The modulus of elasticity of the FRP laminates was 210 GPa with a standard deviation of 5.1 GPa. The CFRP textile consisted of four tows of carbon fibre in the longitudinal direction bonded with transverse fibres as shown in Fig. 2. The cross-sectional dimensions for each tow were  $2.2 \times 1.5$  mm, resulting in a cross-sectional area of  $3.25 \text{ mm}^2$ . The textile was provided by Fortress Stabilisation Systems in 320 mm long strips. The average tensile strength of each fibre tow was 2160 MPa with a standard deviation of 25 MPa based on 5 test samples. The epoxy resin used in the FRP joint tests was Master Brace ADH 4000. The properties of the FRP used are summarised in Table 2.

The FRP-to-concrete joint tests were prepared by cutting 5 mm wide  $\times$  22 mm deep grooves in a series of concrete prisms. The concrete used in the prisms consisted of ordinary portland cement, sand and aggregate at a mix ratio of 1:1.67:2.73. The concrete strength of the prisms was determined by testing 5 concrete cylinders 100 mm in diameter and 200 mm high, in accordance with AS 1379 [30]. Based on the test results, the average compressive strength was 32 MPa and the standard deviation was 2.7 MPa. The average concrete tensile strength was 5.3 MPa and the standard deviation was 0.52 MPa.

### 2.1.2. Sample preparation

For the GO-reinforced CBA samples, the dry GO powder was initially placed in 400 ml water, hand-stirred for two minutes and then sonicated for a further two minutes at room temperature using a probe sonicator at 1000 W. The sonicated GO solution was allowed to stand for 5 min and then added to the remaining water to the quantity required for the specific mix. The resulting water solution which included the sonicated GO was then mixed with the cement for 3 min using a Hobart mixer. The GNP samples were prepared in a similar manner to the GO samples. However, since early trials had found that sonication of the GNP paste was

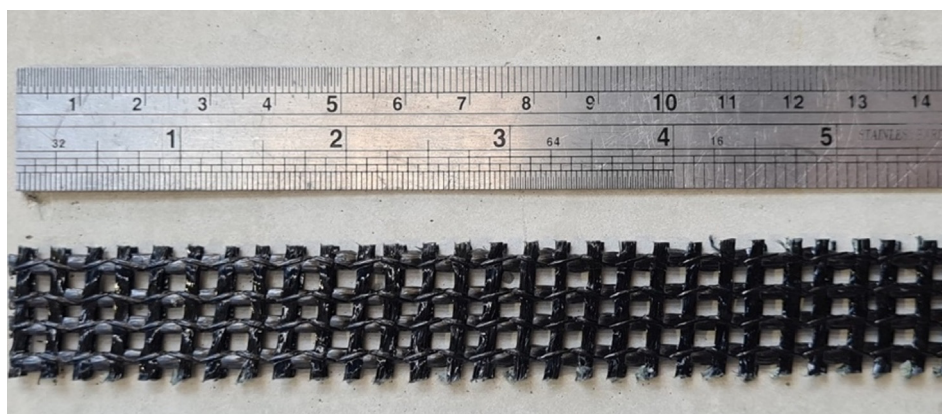
not required to achieve consistent results, no sonication was used in this case. The NS samples were prepared by initially hand-stirring the colloidal silica with the total water content required for each mix. The resulting mix which contained the NS was then mixed with the cement and mixed for a further 3 min in a concrete cake mixer. Superplasticizer was added to achieve the required flowability. This was particularly the case for the NS mixes. In all cases, after the mixing was completed, the cement-based mixes were poured into moulds and vibrated for 10 s. At a time approximating 24 h after casting, the samples were demoulded and then cured in water for 27 days prior to testing.

All mixes used a 0.4 water-to-cement ratio and the quantities of cement, water, GO, GNP, NS and superplasticiser used in each mix are presented in Table 3.

### 2.1.3. Testing methods

Flowability testing was conducted on fresh CBA mixes according to ASTM C1437 [31] to investigate the effect of nano-materials on adhesive workability. The cone dimensions were 70 mm and 100 mm for the top side and base respectively, and the height was 50 mm. After filling the cone with the fresh mix, the cone was lifted, and the table was dropped 25 times in 15 mins. Flowability was then measured as the percentage of the increase in base diameter compared with the original cone base. Proper flowability of the CBA is essential for NSM strengthening, as the groove width is approximately 5 to 7 mm. For low workability mixes where NS was used, the flowability of the CBA was adjusted by adding a water-reducing agent. The compressive and flexural strengths were measured according to BS EN 196-1 [32]. The prism dimensions were  $40 \times 40 \times 160$  mm for the flexural strength tests. The compressive strength of CBA was determined from the fractured flexural prism samples. Care was taken to ensure that each prism sample was uncracked prior to testing.

For the chloride penetration tests, three samples were prepared to investigate the chloride depth in each mix,  $40 \times 40 \times 160$  mm prisms were submerged in a NaCl solution at 10 percent concentration by weight. At 7, 14 and 28 days, three samples of each mix were broken and the chloride penetration depth was measured



**Fig. 2.** CFRP textile.



**Table 2**  
FRP properties.

| Material      | Elastic modulus (GPa) | Ultimate tensile (MPa) | Ultimate strain % | Width (mm) | Thickness (mm) |
|---------------|-----------------------|------------------------|-------------------|------------|----------------|
| CFRP laminate | 210                   | 3300                   | 1.56              | 20         | 1.4            |
| CFRP textile  | 146                   | 2160                   | 1.48              | 20         | 1.5            |

**Table 3**  
Mix proportions.

| Mix               | Cement (g) | Water (g) | GNP (g) | GO (g) | NS (g) | Superplasticizer (ml) |
|-------------------|------------|-----------|---------|--------|--------|-----------------------|
| Reference         | 3000       | 1200.0    | 0.00    |        | –      | –                     |
| GNP 0.025%        | 3000       | 1198.4    | 2.27    |        | –      | –                     |
| GNP 0.05%         | 3000       | 1196.8    | 4.55    |        | –      | –                     |
| GNP 0.075%        | 3000       | 1195.2    | 6.82    |        | –      | –                     |
| GO 0.025%         | 3000       | 1200.0    |         | 0.75   | –      | –                     |
| GO 0.05%          | 3000       | 1200.0    |         | 1.50   | –      | –                     |
| GO 0.1%           | 3000       | 1200.0    |         | 3.00   | –      | 1                     |
| NS 1%             | 3000       | 1015.7    | –       |        | 30     | 2                     |
| NS 2%             | 3000       | 831.4     | –       |        | 60     | 5                     |
| NS 3%             | 3000       | 647.1     | –       |        | 90     | 7                     |
| NS 3% - GNP 0.05% | 3000       | 642.6     | 1.50    |        | 90     | 7                     |
| NS 2% - GO 0.05%  | 3000       | 831.4     |         | 1.50   | 60     | 7                     |
| NS 3% - GO 0.05%  | 3000       | 647.1     |         | 1.50   | 90     | 8                     |

by spraying  $\text{AgNO}_3$  0.1 mol/L solution onto the freshly-broken samples. The chemical reaction led to the development of white and brown zones with obvious boundaries. The depth of the white area is the chloride depth and the brown area represents the chloride-free zone. After spraying with silver nitrate, an immediate reaction is observed when the chloride zone and free chloride zone colours change to white and light yellow-brown, respectively [33].

Single-lap shear specimens were used to assess the bond between the CFRP and the cementitious mixes and epoxy. Prisms 75 mm deep  $\times$  75 mm wide  $\times$  280 mm long were prepared using the respective concrete mix. After the prisms had cured in water for a period of 28 days, a 6 mm wide  $\times$  20 mm deep groove was cut into the top face of each prism for the introduction of FRP into the groove in a NSM application. Although, the CBA mixed were cured in water for 28 days prior to testing, we have preliminary results which suggest that that wet curing for a period of 7 days is sufficient to achieve approximately 70% of the CBA strength. This will need to be taken into account in design when determining the time required to achieve full working strength using CBA materials.

A bond length of 180 mm was used in each case. The bond length commenced 50 mm away from the loading face, as depicted in Fig. 3. After the groove was cut, compressed air was used to clean the groove prior to application. Application involved filling the groove with the respective bonding material (CBA or epoxy) and then placing the FRP strips in the groove. Excess adhesive expelled after the FRP was pressed into place was removed. Finally, the front face of each specimen was painted white and a black high-contrast speckle pattern was applied along the FRP strip for data acquisition using a digital image correlation (DIC) system.

A rigid test rig was constructed to secure the specimens, both vertically and laterally, during testing. The test set-up was based on a standard near-end supported direct shear test configuration and is depicted in f. The top steel loading plate of the test rig was 15 mm thick and restrained by four 12 mm diameter steel threaded rods secured to the 30 mm thick steel base with the specimen in between, as shown in Figs. 4 and 5. The alignment of all specimens was carefully adjusted prior to load application to ensure that the load was applied concentrically. The load was applied using a 100 kN Universal testing machine under displacement

control. The displacement was applied at a rate of 0.1 mm/min. The data acquisition rate was set at 2 Hz to coincide with the measurement frequency of the DIC system, in order to enable precise matching of the data.

The FRP laminate had a free length of 100 mm to allow for loading and gripping. Prior to gripping the FRP, two 50 mm  $\times$  50 mm  $\times$  2 mm aluminium plates were adhered to the laminate gripping area using Araldite epoxy, to ensure that the FRP was not damaged during gripping.

### 3. Results and discussion

The following section outlines the results of all tests conducted in this study. The results are presented in terms of CBA flowability, mechanical properties, chloride penetration (durability), and bond strength.

#### 3.1. Flowability

In general, the addition of nano-materials such as GO, GNP and NS increases the surface area of particles available to react with the water content of the mix, which adversely affects workability. The results indicated no noticeable reduction in workability as a result of the addition of GO and GNP mixes in the quantities at which they were introduced. However, the addition of NS was found to significantly reduce the workability of the mix, which necessitated the addition of superplasticiser to maintain an acceptable level of workability. More specifically, the addition of 3% of NS to the mix resulted in a 42% reduction in the paste diameter for the flowability test. The amount of superplasticiser added to each mix is presented in Table 3.

#### 3.2. Mechanical properties of CBA

The mechanical properties of each CBA mixture were assessed in terms of compressive strength and flexural tensile strength in accordance with the test methods outlined in BS EN 196-1 [32].

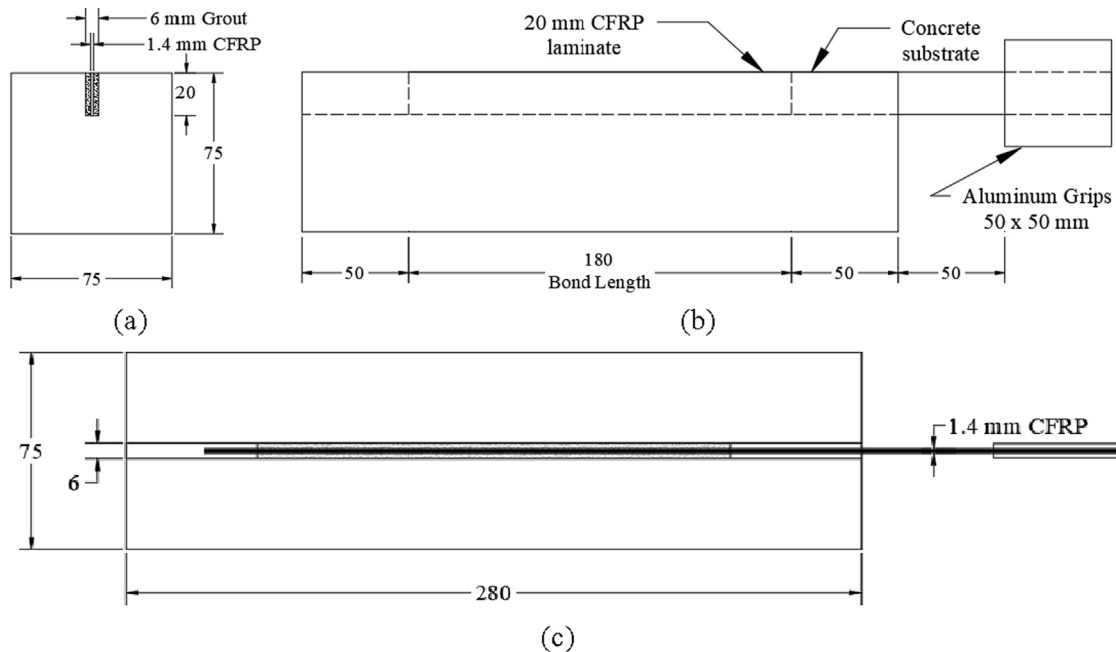


Fig. 3. (a) Cross section; (b) side view; (c) plan view of pull-out test samples.

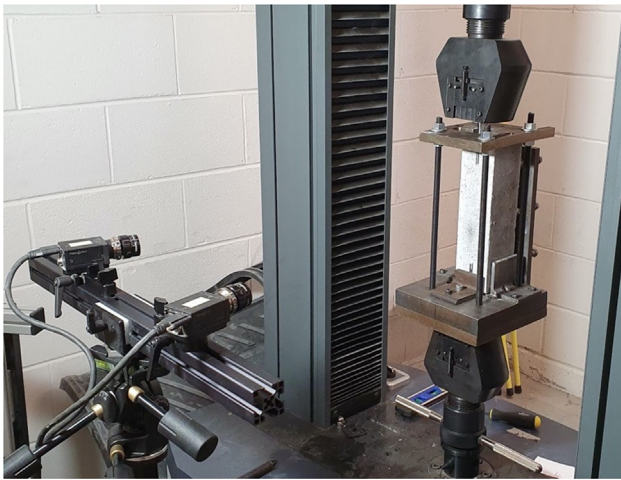


Fig. 4. Test set-up for single shear test.

For each mix, 6 samples were tested at 7 and 28 days. The compressive and flexural strength values for all mixes tested at 7 and 28 days are summarised in Tables 4 and 5, respectively.

The results in Table 4 suggest that the reference mix which contained a water/cement (w/c) ratio of 0.4 and no nano-materials had a compressive strength of 51.5 MPa and 65.9 MPa at 7 and 28 days, respectively. The tensile strength of the reference mix was 9 MPa and 9.4 MPa at 7 and 28 days, respectively.

The compressive and flexural strengths at 7 and 28 days for different GNP contents are summarised in Fig. 6. The results show no discernible improvement in CBA compressive strength as a result of the addition of GNP to the mix. These results correlate with those obtained by other researchers [34], who also noted that any enhancements in the mechanical properties of cementation materials due to the addition of GNP were confined to tensile strength only. Alatawna et al. [35] reported an enhancement of

43% in the flexural strength of cement composite at 0.1% of GNP and 30% improvement in compressive strength at the same loading. In addition, Zohhadi et al. [36] concluded that the addition of GNP to cement mortar improves flexural and compressive strength by 39% and 28%, respectively. The addition of GNP increases the flexural tensile strength of the CBA in proportion to the quantity of GNP added. As a result, the addition of 0.075% of GNP by weight to the mix increases the flexural tensile strength from 9 MPa to 11.3 MPa, corresponding to a 21% improvement. GNP contents greater than 0.075% were also investigated. However, decreases in both compressive and flexural tensile strength were observed and the results were inconsistent. This was attributed to clustering of the GNP at higher concentrations causing agglomeration, and the difficulty of achieving uniform dispersion.

A summary of the compressive and flexural tensile strengths for CBA mixes enhanced by the addition of GO at 7 and 28 days is presented in Fig. 7. The results show a trend in flexural and compressive strengths up to 0.05% of GO. It can be observed that doses higher than 0.05% of GO have a negative effect on mechanical properties. The results suggest an optimal GO content of 0.05%, which resulted in an increase in 28 day compressive strength of 6% and flexural tensile strength of 24%. Similar research results were reported by An et al. [19], who concluded that 0.05% of GO improved the compressive and flexural strengths of cement paste by 25% and 37%, respectively.

Fig. 8 summarises the influence of the addition of NS on compressive and flexural tensile strength. In contrast to the GNP and GO samples, which showed no or only marginal improvements in compressive strength, the addition of NS up to 3% increased the compressive strength of the CBA from 66 MPa to 85 MPa, a 29% increase. Further, the flexural tensile strength increased by 37% with respect to the reference mix up to a maximum value of 12.8 MPa when 3% of NS was added to the mix. The 3% NS content was considered to be optimal, as any further increase in concentration resulted in low workability, even with the addition of larger quantities of superplasticiser.

The NS tends to fill the pores in the CBA and improves the pozzolanic reaction, resulting in a denser microstructure and higher

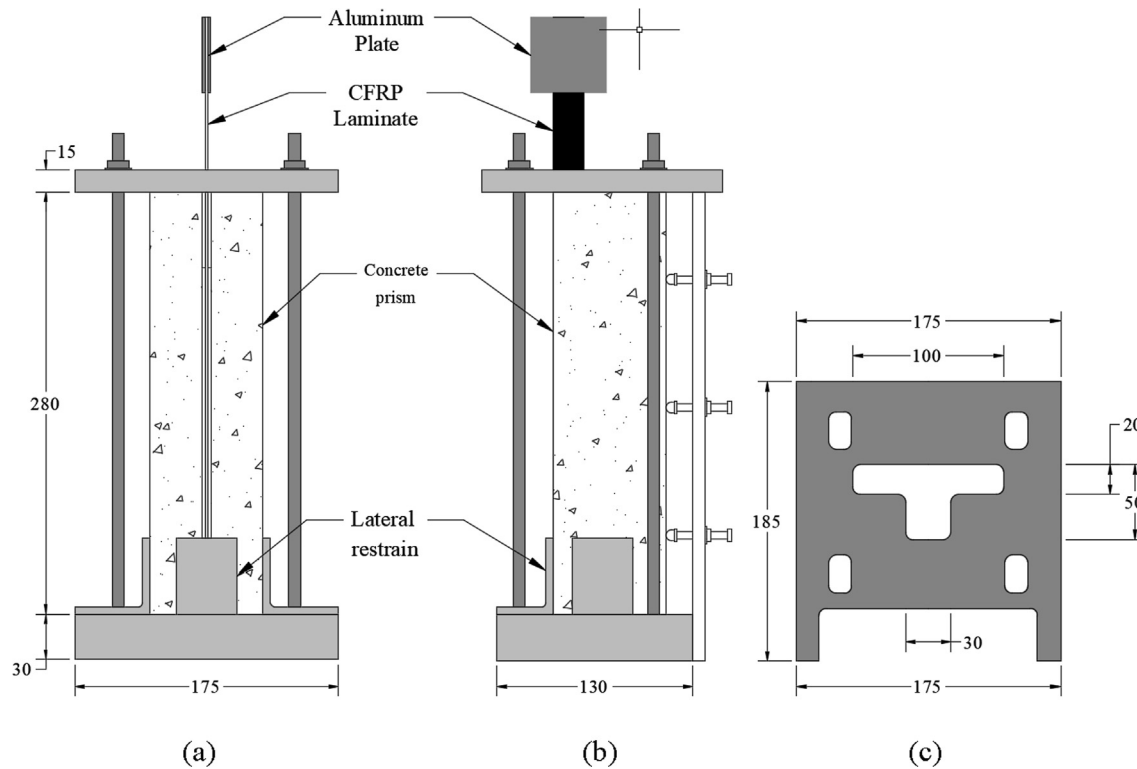


Fig. 5. (a) Front view; (b) side view; (c) loading plate plan for single shear test.

**Table 4**  
Mechanical properties of CBA mixes at 7 days.

| Mix               | Compressive strength (MPa) |      |      |      |         | Flexural strength (MPa) |      |      |      |         |
|-------------------|----------------------------|------|------|------|---------|-------------------------|------|------|------|---------|
|                   | S1                         | S2   | S3   | Avg. | Diff. % | S1                      | S2   | S3   | Avg. | Diff. % |
| Reference         | 50.5                       | 51.2 | 51.5 | 51.1 | –       | 9.0                     | 9.3  | 9.1  | 9.1  | –       |
| GNP 0.025%        | 55.1                       | 56.1 | 53.6 | 54.9 | 8%      | 9.5                     | 10.5 | 9.5  | 9.8  | 8%      |
| GNP 0.05%         | 52.5                       | 41.5 | 56.0 | 50.0 | –2%     | 9.9                     | 10.6 | 10.7 | 10.4 | 14%     |
| GNP 0.075%        | 55.0                       | 56.4 | 56.5 | 56.0 | 10%     | 11.0                    | 11.3 | 10.1 | 10.8 | 18%     |
| GO 0.025%         | 57.3                       | 58.1 | 57.5 | 57.6 | 13%     | 10.1                    | 11.1 | 9.5  | 10.2 | 12%     |
| GO 0.05%          | 58.2                       | 58.0 | 57.9 | 58.0 | 14%     | 10.1                    | 10.3 | 10.5 | 10.3 | 13%     |
| GO 0.1%           | 58.1                       | 57.2 | 57.7 | 57.7 | 13%     | 9.8                     | 9.7  | 12.0 | 10.5 | 15%     |
| NS 1%             | 64.2                       | 57.8 | 61.6 | 61.2 | 20%     | 10.8                    | 11.4 | 10.4 | 10.9 | 19%     |
| NS 2%             | 67.7                       | 62.6 | 68.8 | 66.4 | 30%     | 11.3                    | 10.7 | 11.4 | 11.1 | 22%     |
| NS 3%             | 70.9                       | 71.1 | 70.2 | 70.7 | 38%     | 11.6                    | 11.3 | 11.3 | 11.4 | 25%     |
| NS 3% - GNP 0.05% | 61.4                       | 66.8 | 62.8 | 63.7 | 25%     | 7.8                     | 12.5 | 11.8 | 10.7 | 17%     |
| NS 3% - GO 0.05%  | 72.0                       | 71.0 | 74.8 | 72.6 | 42%     | 11.0                    | 11.8 | 11.5 | 11.4 | 25%     |

**Table 5**  
Mechanical properties of CBA mixes at 28 days.

| Mix               | Compressive strength (MPa) |      |      |      |         | Flexural strength (MPa) |      |      |      |         |
|-------------------|----------------------------|------|------|------|---------|-------------------------|------|------|------|---------|
|                   | S1                         | S2   | S3   | Avg. | Diff. % | S1                      | S2   | S3   | Avg. | Diff. % |
| Reference         | 66.5                       | 65.0 | 66.2 | 65.9 | –       | 9.4                     | 9.3  | 9.4  | 9.4  | –       |
| GNP 0.025%        | 64.0                       | 66.0 | 65.0 | 65.0 | –1%     | 10.9                    | 10.7 | 10.5 | 10.7 | 14%     |
| GNP 0.05%         | 63.0                       | 66.0 | 65.0 | 64.7 | –2%     | 11.0                    | 11.4 | 10.0 | 10.8 | 15%     |
| GNP 0.075%        | 62.5                       | 63.3 | 63.0 | 62.9 | –5%     | 11.1                    | 11.5 | 11.4 | 11.3 | 21%     |
| GO 0.025%         | 66.0                       | 67.0 | 67.8 | 67.0 | 2%      | 10.6                    | 11.6 | 11.9 | 11.4 | 21%     |
| GO 0.05%          | 71.0                       | 72.0 | 70.0 | 70.0 | 8%      | 11.3                    | 12.0 | 11.5 | 11.6 | 24%     |
| GO 0.1%           | 62.8                       | 60.2 | 61.5 | 61.5 | –7%     | 11.8                    | 11.7 | 11.0 | 11.5 | 23%     |
| NS 1%             | 74.0                       | 73.5 | 73.7 | 73.7 | 12%     | 13.0                    | 11.6 | 12.1 | 12.2 | 31%     |
| NS 2%             | 80.9                       | 79.0 | 80.1 | 80.0 | 21%     | 12.8                    | 11.9 | 12.5 | 12.4 | 32%     |
| NS 3%             | 84.2                       | 85.3 | 86.0 | 85.2 | 29%     | 12.0                    | 14.6 | 11.9 | 12.8 | 37%     |
| NS 3% - GNP 0.05% | 78.1                       | 79.0 | 79.5 | 78.9 | 20%     | 12.0                    | 11.5 | 11.6 | 11.7 | 25%     |
| NS 3% - GO 0.05%  | 86.5                       | 84.1 | 83.0 | 84.5 | 28%     | 14.0                    | 13.5 | 13.7 | 13.7 | 47%     |

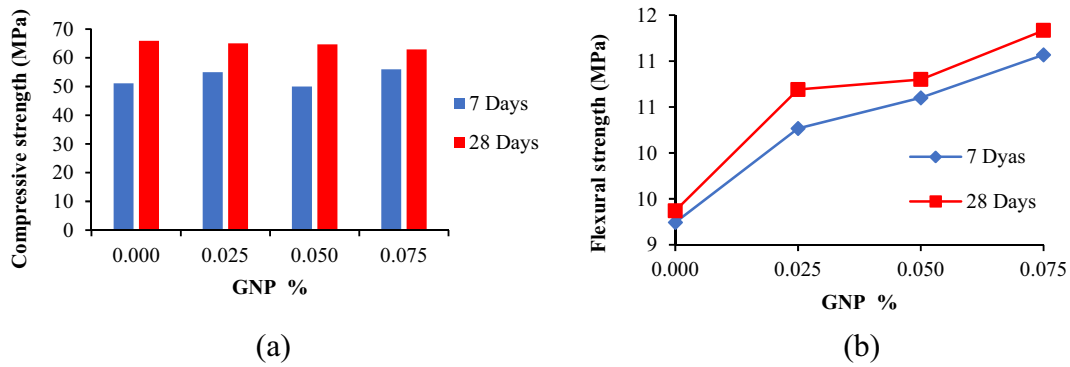


Fig. 6. Summary of compressive and flexural strengths at 7 and 28 days for different GNP quantities; (a) Compressive strength; (b) Flexural tensile strength.

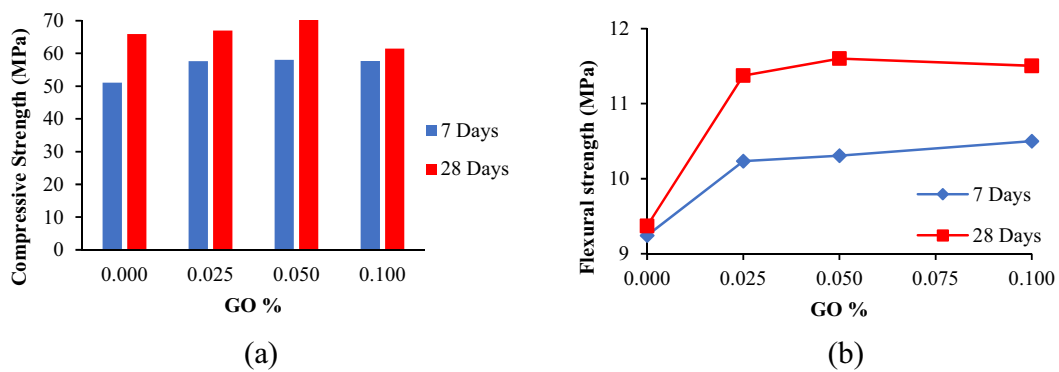


Fig. 7. Summary of compressive and flexural strengths at 7 and 28 days for different GO quantities; (a) Compressive strength; (b) Flexural tensile strength.

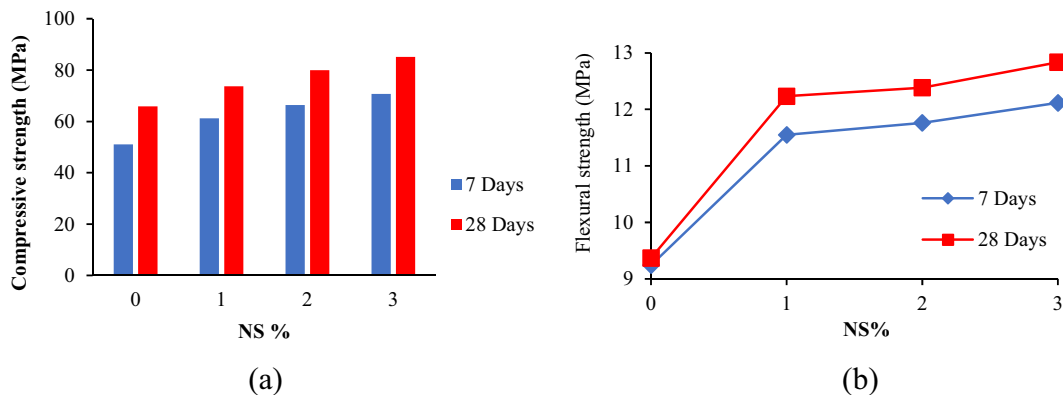


Fig. 8. Summary of compressive and flexural strengths at 7 and 28 days for different NS quantities; (a) Compressive strength; (b) Flexural tensile strength.

mechanical properties [16,37]. This was also confirmed by Said et al. [38], who suggested that the use of NS refines the concrete microstructure because of the improvement of the chemical reaction and the nano-filler effect of the NS. The experimental results showed that although doses of NS higher than 3% improve the mechanical properties, workability is compromised, and higher dosages of water and superplasticiser are required.

Introducing GO and GNP to NS mixes resulted in additional improvement to the mechanical properties. NS 3%-GO 0.05% and NS 3%-GNP 0.05% showed slightly better performance than NS 3%. Moreover, the NS 3%-GO 0.05% mix had the best performance

of all mixes, as the compressive strength improved by 28% and the flexural strength improved by 47% based on the average of 6 samples at 28 days, as shown in Table 5. The improvement in the mechanical properties can be attributed to a better pozzolanic reaction and the nano-filler effect of the nano-materials, as discussed previously.

### 3.3. Chloride penetration depth

The chloride penetration of each CBA mix was assessed in order to provide information on the porosity and durability of each mix.



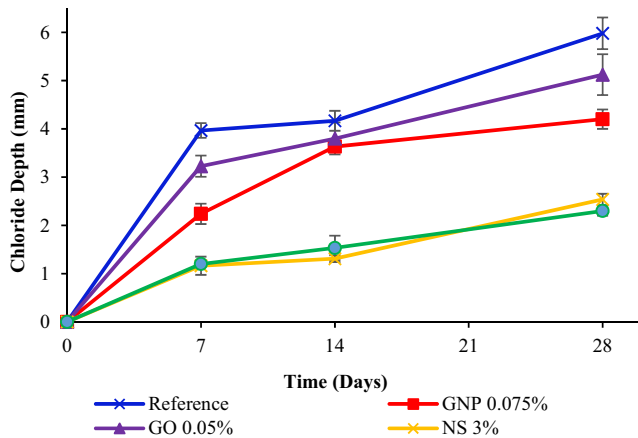


Fig. 9. Chloride penetration depths in mm for different mixes at different ages.

Reduced porosity results in lower chloride penetration and a greater degree of protection to steel reinforcement, which have implications for a variety of applications. The depth of the chloride penetration zone was measured using Verniers combined with scaled images in CAD. The average chloride penetration depth for three sample of each mix at 7, 14 and 28 days are presented in Fig. 9, and images of the chloride penetration zone at 28 days are presented in Fig. 10. The results indicate that the addition of 0.075% of GNP to the reference mix reduces the chloride penetration depth by 26%. In contrast, the addition of 0.05% of GO reduces the chloride penetration depth by 10% and is less effective than GNP. The greatest reduction in chloride penetration depth was observed for the NS specimens, as 3% of NS reduced the chloride

penetration by 61%. The combination of 3% NS with 0.05% GNP did little to enhance the chloride penetration depth with respect to pure NS and a 54% reduction was achieved.

Du and Pang [39] reported that the addition of 2.5% of GNP reduces the chloride penetration depth by 31%. Similarly, Zhang and Li [40] concluded that the addition of 3% of NS improves the chloride penetration depth by 10.5%. Further research by Jalal et al. [41] found that the chloride penetration depth decreased by 52% when 2% of NS was added. Resistance to chloride penetration is attributed to the refinement of the pore structure in the cement paste [7].

### 3.4. Pull-out test results

After characterising the mechanical properties of the respective mixes in terms of compressive and flexural tensile strengths, it was important to determine the effectiveness of the CBA materials when used as a bonding agent between FRP laminates and concrete prisms, in order to assess the suitability of the CBA mixes in NSM FRP strengthening applications as a replacement for epoxy resin. This section presents the results of the FRP-to-concrete near-end supported direct shear tests. All specimens were tested at 28 days from pouring the CBA into the grooves. A summary of the force required to fail each specimen and the maximum FRP strain achieved and the average bond stresses between the FRP and the CBA material in each case is presented in

Table 6. It should be noted that the maximum strain in the FRP was measured at one location along the FRP free length. The failure modes for the respective CBA mixes are presented in Fig. 11.

Generally, failure of all CBA specimens occurred by FRP debonding which occurred at either (1) the interface between the FRP and the CBA; or (2) the interface between the surface of the concrete groove and the CBA. Of the two interface debonding failure planes,

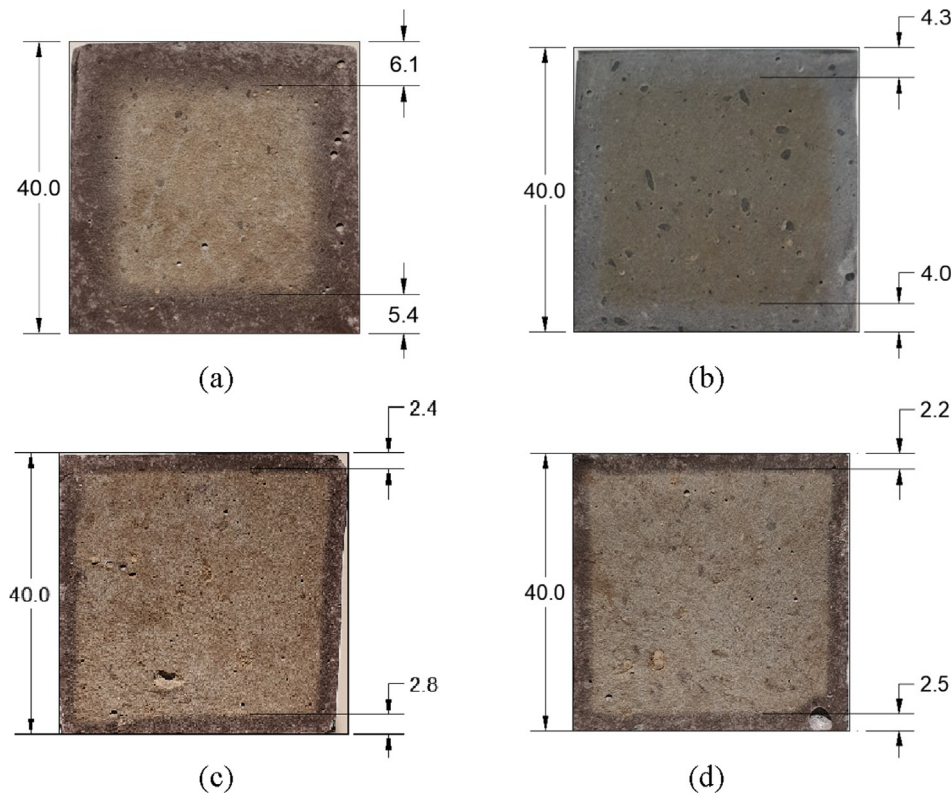


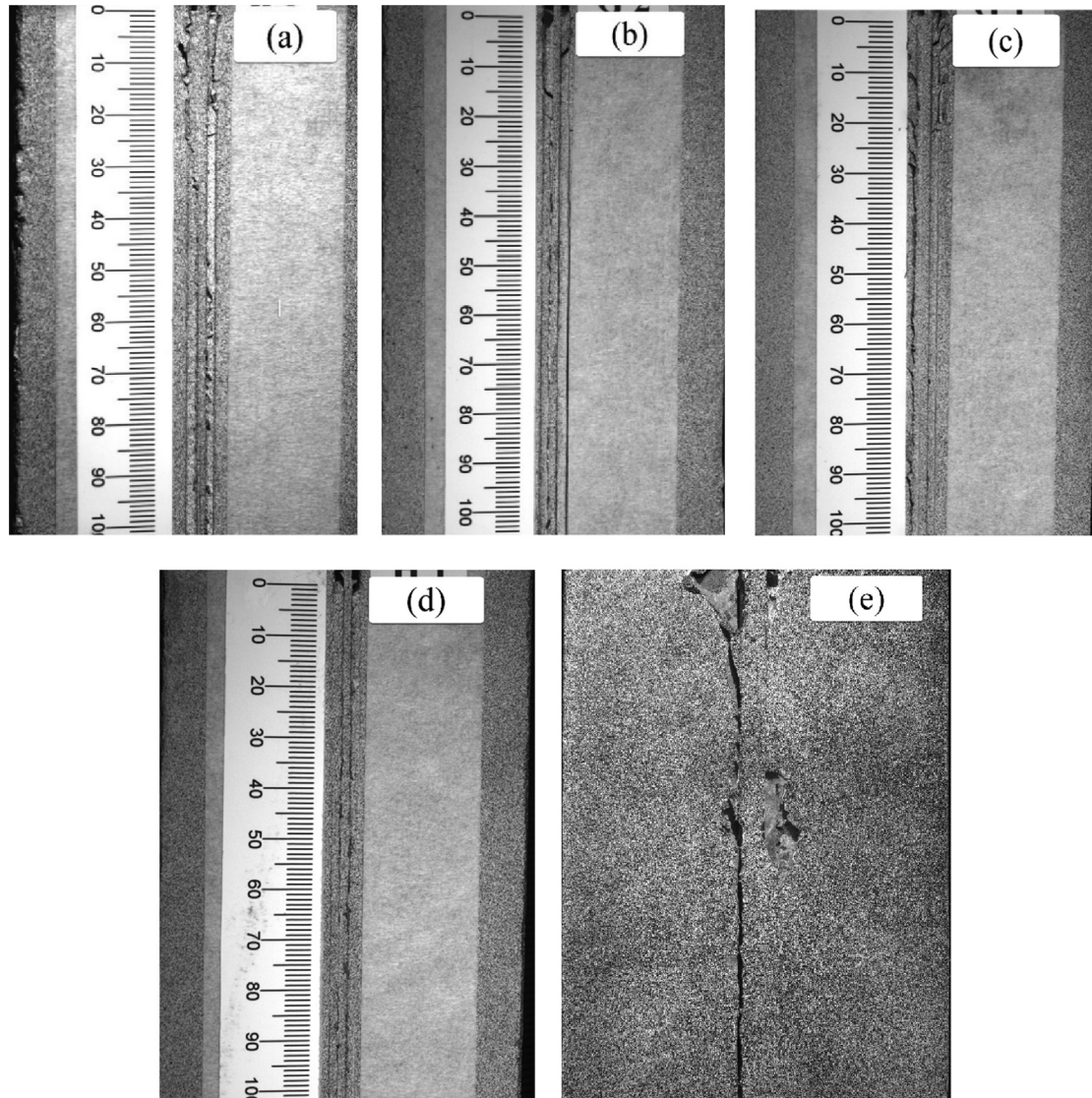
Fig. 10. Chloride penetration depths in mm for different mixes at 28 days; (a) Reference mix; (b) GNP 0.075%; (c) NS 3%; (d) GNP 0.05% - NS 3%.

**Table 6**

Summary of CFRP laminate bond assessment results for different CBA mixes.

| Adhesive              | Pull-out force (kN) |      |      |      | Diff. (%) | Max. $\mu$ strain | Fracture Energy N/mm | Failure Mode* |
|-----------------------|---------------------|------|------|------|-----------|-------------------|----------------------|---------------|
|                       | S1                  | S2   | S3   | Avg. |           |                   |                      |               |
| Reference             | 17.2                | 16.7 | 17.7 | 17.2 | –         | 4095              | 0.122                | CLI           |
| GNP 0.075%            | 21.8                | 22.3 | 21.2 | 21.8 | 27        | 5190              | 0.339                | CCI, CLI      |
| GO 0.05%              | 18.8                | 19.5 | 21.1 | 19.5 | 13        | 4645              | 0.267                | CCI, CLI      |
| NS 3%                 | 12.6                | 11.9 | 12.1 | 12.2 | –12       | 3620              | 0.104                | CLI           |
| NS 3% - GO 0.05%      | 12.4                | 13.0 | 12.1 | 12.5 | –10       | 3255              | 0.088                | CLI           |
| Master Brace ADH 4000 | 47.1                | 43.7 | 44.4 | 45.0 | 162       | 10,200            | 1.577                | CLI           |

\*Note: CLI: CBA/laminate interface, CCI: concrete/CBA interface.

**Fig. 11.** Failure modes for different mixes (a) reference mix; (b) GO 0.05%; (c) G0.075%; (d) NS 3% - GO 0.05% Mix; (e) Master Brace ADH 4000.

failure between the FRP and the CBA interface was dominant. This is clearly noted in Table 6. For the purposes of comparison, Master Brace ADH 4000 was also used as a bonding agent and in this case, failure of all Master Brace ADH 4000-bonded specimens occurred at the interface between the concrete groove and the epoxy. The fracture energy for all mixes used was calculated by determining the area under the bond slip curves in Fig. 15. These fracture energy values are summarised in Table 6. Generally, the addition

of GNP 0.075% to the control CBA mix increased its fracture energy by 178%. However, the epoxy resin used still had a fracture energy which was 359% higher than the GNP 0.075% mix. Greater fracture energies were a result of higher peak bond stresses and corresponding slip values before failure in addition to greater post peak deformation prior to failure.

An examination of the results for the pull-out forces required to fail the FRP-to-concrete joints presented in Table 6 reveals that



very consistent results were achieved for the three samples tested for each mix (S1-S3). The reference mix comprised standard unmodified cement paste and an average pull out force of 17.2 kN was required to fail the joint. The addition of 0.075% GNP improved upon this, as the pull-out force increased to 21.8 kN, a 27% improvement. This corresponded to an increase in FRP strain prior to debonding from 4095  $\mu\epsilon$  to 5190  $\mu\epsilon$ . A lower level of bond strength enhancement of 13% was observed when 0.05% GO was added. None of the specimens tested which contained pure NS and a combination of NS and GO showed any improvement in bond performance. In fact, specimens which contained NS were observed to fail at a 10% lower pull-out force with respect to the reference specimen. The specimens for which the FRP was bonded using Master Brace ADH 4000 had the highest pull-out force for the FRP and failed at an average of 45 kN and maximum FRP laminate strain of 10,200  $\mu\epsilon$ . Although the GNP-modified CBA showed a marked improvement in bond performance compared with the reference mix, the FRP strain to cause debonding was still only 33% of the FRP rupture strain, and there was still further room for improvement. It was agreed that an improved mechanical interlocking between the FRP and CBA could result in better FRP utilization and a higher debonding force. As a result, carbon fibre textile was trialled in place of the FRP laminate.

The textile had an open weave, which meant that the CBA material could penetrate the FRP strip, thereby mitigating the failure mode which occurred between the interface of the FRP laminate and the CBA. Table 7 depicts the results of pull-out tests using FRP textile and different CBA mixes. All the modified CBA mixes performed better in the pull-out tests than the reference mix by approximately 27%. It can be concluded that all the modified mixes achieved full utilisation of the FRP textile and as a result, ruptured at their ultimate capacity. Due to full utilisation of the FRP textile being achieved when using a single textile strip, inserting two textile strips into a single groove was also trialled, as this would reduce the number of grooves and the costs in real applications if successful.

When two FRP textile strips in a single groove were used, the results showed a significant increase in the maximum pull-out force when bonded with the GNP mix. The average pull-out force for double FRP textile for three specimens was 37.63 kN, a 73% improvement over the reference test. The failure mode was governed by concrete splitting around the textile and subsequent debonding at the CBA and the concrete prism interface, as shown in Fig. 12.

### 3.5. Strain measurements

In addition to the force captured during testing of the FRP-to-concrete joint samples, DIC was used to measure the strain distribution along the length of the FRP laminate bond line during loading as shown in Fig. 13. These measurements provide insights into

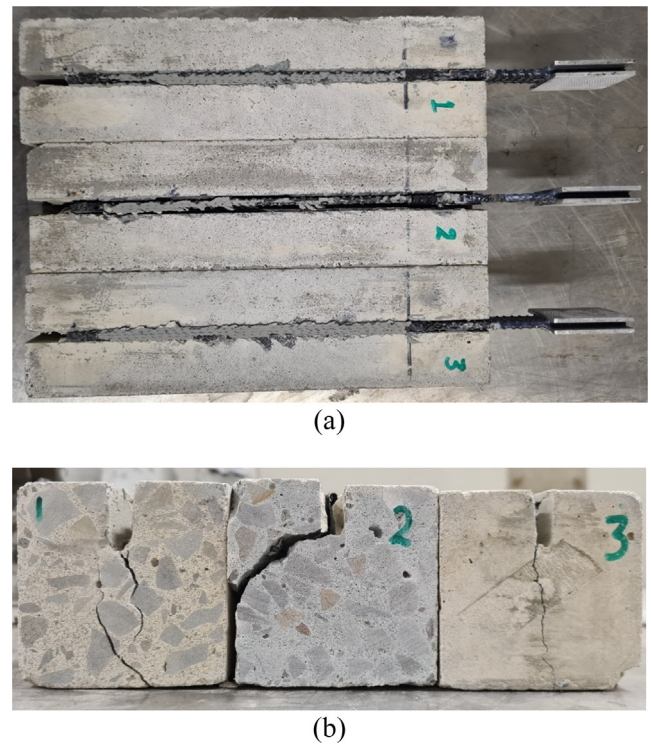


Fig. 12. Failure of double textile pull-out tests (a) top view; (b) side view.

the following phenomena: (1) the effective bond length of the FRP can be determined using CBA and a comparison made with epoxy resin; (2) the strain distribution along the bond line can be used to derive bond-slip formulations for the different CBA mixes. This will provide further insights into the bond properties between the FRP and CBA for use in future numerical simulations or in the development of design formulations. The strain maps for GNP 0.075% mix CBA and Master Brace ADH 4000 can be seen in figure 13 at different loading stages. Using flexible adhesive resulted in smoother stress distribution along the bond length compared to the CBA adhesive. Similar findings were reported by other researchers when comparing flexible and stiff adhesives [3].

The strain distributions along the bond length for different mixes under different loading levels can be seen in Fig. 14. The strain reading at the beginning of the bond distance corresponds to the strain reading at the free length.

An examination of the strain distributions presented in Fig. 14 reveals that the specimens in which the FRP was bonded using epoxy resin required a shorter bond length to develop a certain force in the FRP. For instance, in the case of Master Brace ADH 4000, a bond length of approximately 100 mm was required to

Table 7  
Summary of CFRP textile bond assessment results for different CBA mixes.

| Mix              | Pull-out force (kN) |       |      |      | Diff. (%) | Max. $\mu$ strain | Failure* mode |
|------------------|---------------------|-------|------|------|-----------|-------------------|---------------|
|                  | S1                  | S2    | S3   | Avg. |           |                   |               |
| Reference        | 21.8                | 22.2  | 21.3 | 21.8 | –         | 11,468            | CLI           |
| NS 3%            | 27.6                | 27.9  | 28.0 | 27.8 | 28%       | 14,680            | TEX           |
| GO 0.05%         | 24.9                | 25.3  | 25.7 | 25.3 | 16%       | 13,332            | CLI, TEX      |
| GNP 0.075%       | 26.0                | 24.6  | 25.9 | 25.5 | 17%       | 13,332            | CLI, TEX      |
| NS 3% - GO 0.05% | 26.7                | 26.10 | 26.6 | 26.5 | 22%       | 13,953            | TEX           |
| GNP 0.075% (2)** | 36.5                | 37.9  | 38.5 | 37.6 | 73%       | 9914              | CONC,         |

\*Note: CLI: CBA/laminate interface; TEX: Rupture of CFRP textile; CONC. Concrete failure.

\*\* Double CFRP textile.

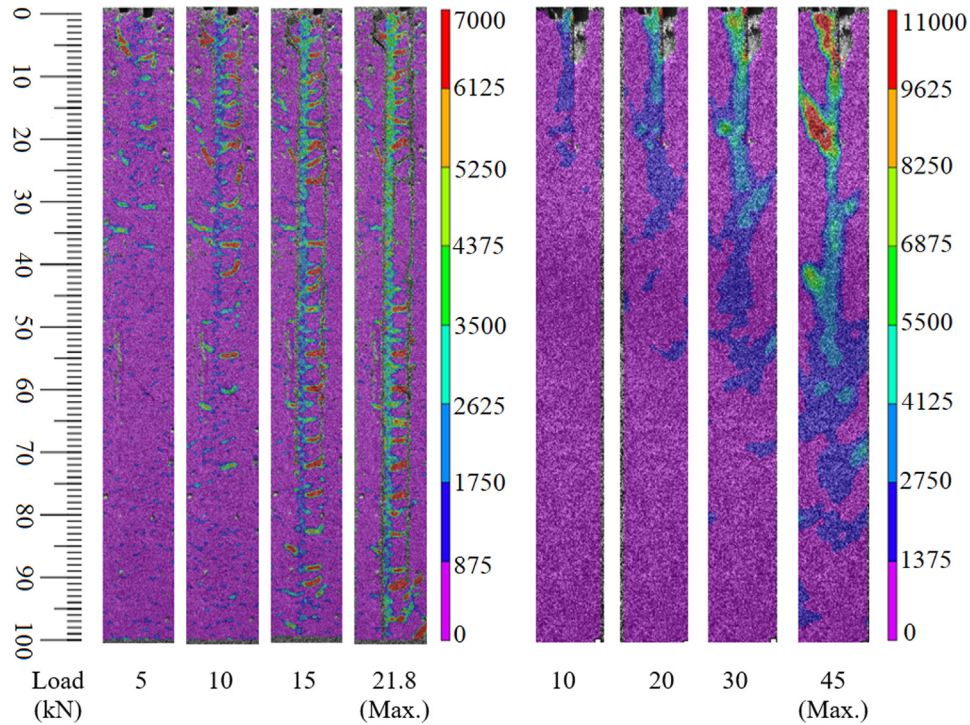


Fig. 13. Strain  $\varepsilon_{yy}$  maps as acquired from DIC system in  $\mu\epsilon$  at different loading (a) GNP 0.075% mix (b) Master Brace ADH 4000.

develop an FRP strain level of 4000  $\mu\epsilon$ . However, in the case of the GNP-modified CBA, a bond length of approximately 125 mm was required, as shown in Fig. 14c. Generally, the effective bond length for CBA mixes is expected to be 25% longer the Master Brace ADH 4000. It should also be noted that damage can be observed in many of the CBA mixes prior to the ultimate joint capacity being reached which can be observed by flattening of the FRP strain along the bond line as opposed to a decrease in FRP strain with distance away from the loaded end in the case of a undamaged bond. This can clearly be observed in Fig. 14c-d. When examining Fig. 14d, it was observed that 7kN the load level which damage commenced for the NS 3% - GO 0.05% mix. This was observed as the strain distribution along the bond line started to flatten for loads above 7 kN. This corresponded to commencement of damage at a load level which was 44% lower than the ultimate capacity of the joint for the NS 3% - GO 0.05% mix. However, damage occurred later in the loading process for the GNP 0.075% mix. This can be observed in Fig. 14c where damage was observed to commence at 15 kN which was 31% lower than the ultimate capacity of the joint. For design using CBA materials, it is important that designers ensure that damage will not be present at the serviceability limit state load case.

### 3.6. Bond-slip relationships

The average bond stress between any subsequent two points along the FRP to CBA bond line can be calculated by dividing the force difference between two points by the interfacial bond area between the same two points. This is expressed in Eq. (1):

$$\tau = \frac{E_f(\varepsilon_{f(i+1)} - \varepsilon_{f(1)})t_f}{2\Delta L} \quad (1)$$

where,  $E_f$  is the elastic modulus of FRP laminate 210 MPa,  $t_f$  is the thickness of FRP laminate 1.4 mm,  $\Delta L$  is the distance between two subsequent points and  $\varepsilon_{f,i+1}$  and  $\varepsilon_{f,i}$  are FRP strains.

The average slip is then calculated as the incremental sum of the FRP extension. This is expressed in Eq. (2):

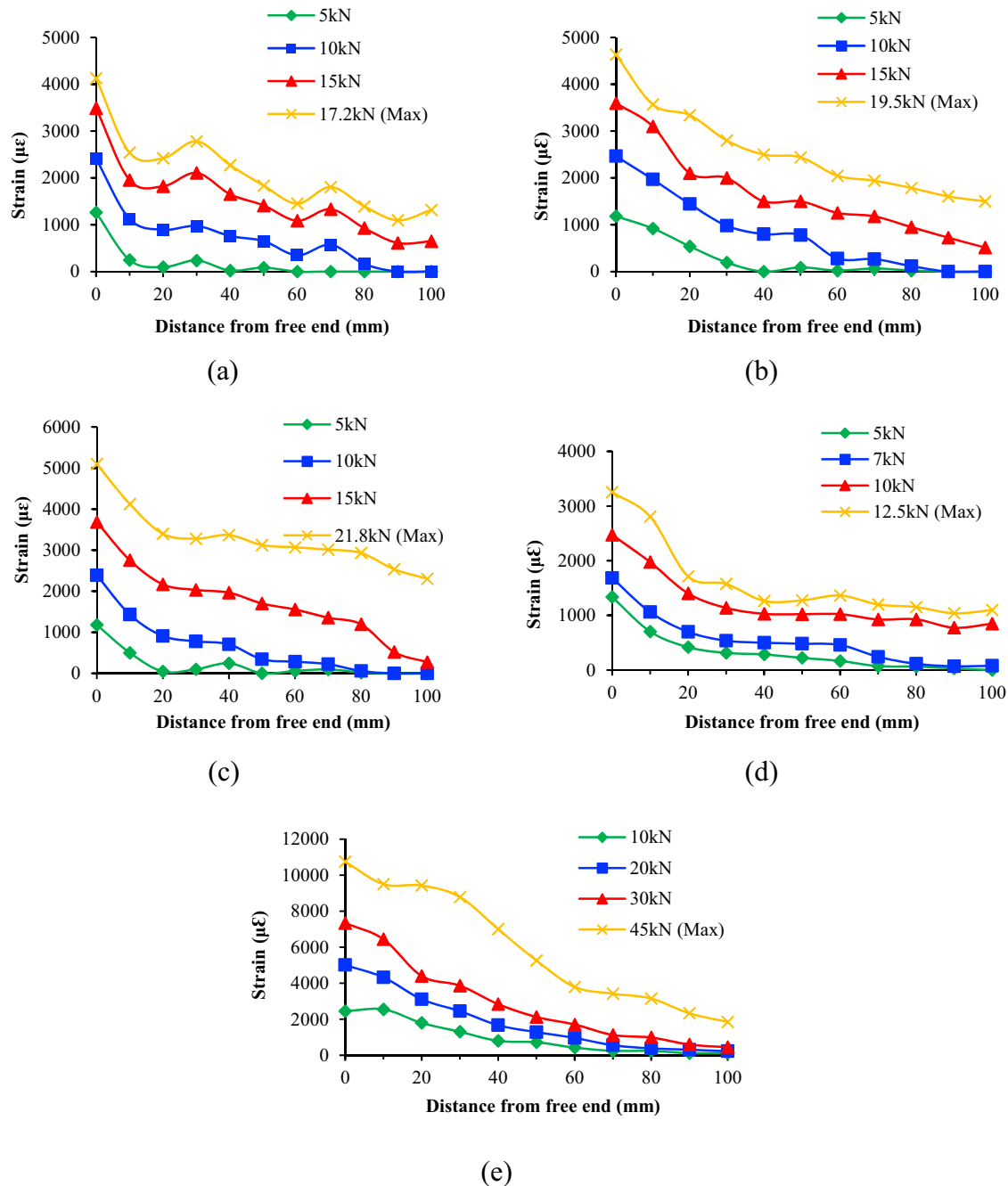
$$s_i = \frac{\varepsilon_{f,i+1} + \varepsilon_{f,i}}{2} \Delta L + s_{i-1} \quad (2)$$

The bond slip relationships between the FRP laminates and all bonding agents explored are depicted in Fig. 15. Note that results for the FRP textiles are not presented here, due to the difficulty of achieving accurate strain measurements along the length of the textile.

The bond-slip results presented in Fig. 15 for the varying mixes indicate that the peak bond stress for the reference mix was 2.4 MPa at a corresponding slip of 0.04 mm. The peak bond stress and corresponding slip for the GNP-modified CBA was 3.57 MPa and 0.085 mm, respectively. This corresponds to a 49% increase in bond strength with respect to the reference mix. The improvement was less pronounced for the GO-modified CBA mix, for which a peak bond stress and a corresponding slip of 3.23 MPa and 0.065 mm, respectively was observed (a 35% increase).

NS-modified CBA achieved the lowest results for the peak bond stresses, achieving a peak bond stress of only 1.47 MPa. On the other hand, the bond stress for Master Brace ADH 4000 was 7 MPa at 0.25 mm slip, which was significantly higher than that of the CBA bonding agents.

The elastic modulus for the CBA mixes were estimated using the homogenization method proposed by Tian and Bian [42] as 20 GPa considering w/c ratio and the cement type. This was confirmed using another model proposed by Wittmann [43] which indicated that the elastic modulus for the hardened cement paste with 0.4 w/c ratio was 21 GPa at 28 days. In both models, the w/c ratio plays the dominant role in predicting the elastic modulus. It should be noted that any influence the nano materials have on the elastic modulus is expected to be minimal and has been ignored in these estimates. On the other hand, the elastic modulus for Master Brace ADH 4000 was 10 GPa as reported by the manufacturer.



**Fig. 14.** Strain distribution along CFRP laminate bond lengths for different bonding agents (a) Reference mix; (b) GO 0.05% mix; (c) GNP 0.075% mix; (d) NS 3% - GO 0.05%; (e) Master Brace ADH 4000.

From examining the bond slip curves derived from the FRP-to-concrete joint tests, it can be observed that both adhesive stiffness and brittleness play a significant role in the corresponding slip achieved at the peak bond stress. In general, all CBA materials exhibited both lower bond stresses and stiffness of the FRP-to-concrete joint when compared to more flexible adhesives such as epoxy resin. The CBA materials used in this study contained a modulus of elasticity which was approximately double that of the epoxy resins used (20 GPa compared with 10 GPa) and this was reflected in the gradient of the ascending branch of the bond slip

curve where the CBA curves in Fig. 15a-d were twice as stiff as the epoxy curve in Fig. 15e. As a result, use of CBA materials in lieu of epoxy resins is expected to change the stress distributions along the FRP-to-concrete bond line and result in higher stress concentrations when compared to more flexible adhesives such as epoxy resins. The influence of this phenomenon on large scale beam strengthening applications will be quantified at a later stage of this study. Although highly deformable adhesives can reduce the stress concentration along the bond line [44,45], they have limited performance when exposed to elevated temperatures.



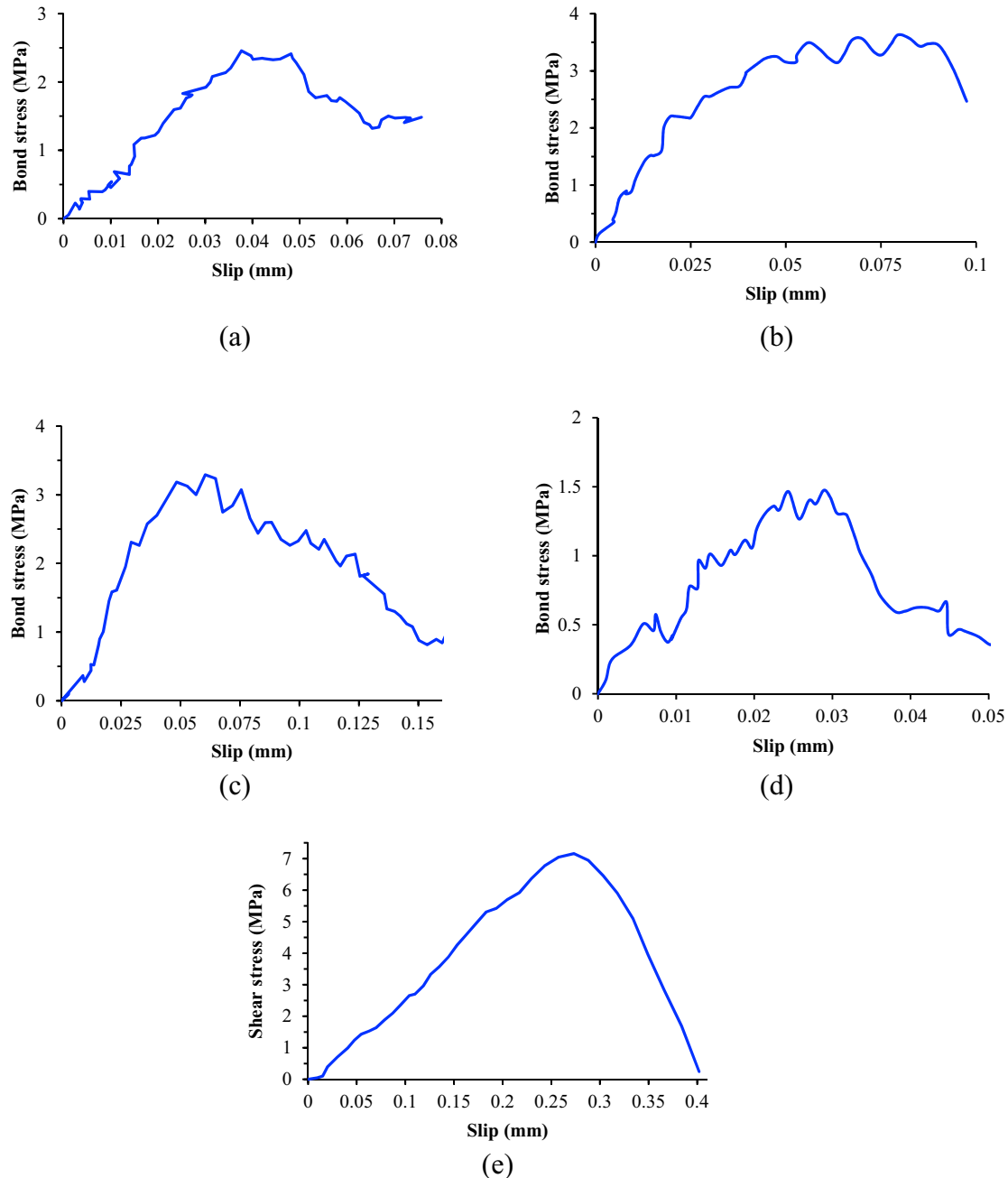


Fig. 15. Bond-slip relationships for different adhesives Reference mix; (b) GO 0.05% mix; (c) GNP 0.075% mix; (d) NS 3% - GO 0.05%; (e) Master Brace ADH 4000.

#### 4. Conclusion

Fire protection of FRP systems of utmost importance. In most countries buildings should have a fire rating of 2 h and in many cases, loss of the FRP in a fire load case would result in collapse of the structure. The use of fire protection and insulation systems are typically 100 mm thick, are costly and impractical or impossible to install in some cases. As a result, the use of CBA is the most promising solution for this problem. This paper has presented the development of several new nano-modified CBA mixes enhanced by the addition of GNP, GO and NS. Although the CBA materials are expected to have a wide potential for applications, they were developed specifically for use as a cement-based adhesive in NSM FRP strengthening applications. When used in this manner, these CBA materials will address

the poor fire performance of organic resins and result in strengthening solutions with superior fire resistance. The mechanical performance of the CBA mixes was evaluated in terms of compressive strength, flexural tensile strength and bond strength, and durability was evaluated using chloride penetration tests. The findings of this research allow the following conclusions to be drawn:

- The addition of 0.075% of GNP to CBA enhances flexural strength by 21% and decreases the chloride penetration depth by 26%. It does not improve compressive strength.
- The addition of 0.05% GO to CBA improves flexural strength by 24% with slight improvement of the chloride penetration depth. No considerable improvement in compressive strength is observed.

- NS-modified CBA increases both mechanical properties and durability. With 3% of NS, compressive and flexural strength increase by 29% and 37% respectively, and chloride penetration is reduced by 61%.
- The hybrid mixes which use a combination of NS and GO show superior performance in compressive and flexural strength. The addition of 3% of NS combined with 1.5% of GO enhances flexural and compressive strength by 57% and 46% respectively, and chloride penetration is reduced by 54%.
- An examination of the bond performance between FRP laminates and CBA for the various mixes revealed that all samples which contained NS achieved a 10% lower pull-out force than the reference mix and were therefore not effective in this case. However, the addition of 0.075% GNP resulted in a 27% improvement in pull-out force and a 49% increase in bond strength with respect to the reference mix.
- Based on all mixes tested, it was noted that the GNP-modified CBA material which contained 0.075% GNP improved both flexural tensile strength and bond strength and is therefore considered to be the most promising candidate for use as a CBA material in NSM FRP applications.
- The use of FRP textile resulted in higher bond properties for all CBA mixes, resulting in rupture of the FRP textile for all modified CBA mixes. When two textile strips were introduced into a single groove, the pullout force increased to 37.65 kN, which was 73% higher than the reference mix for a single laminate strip.
- The fracture energy for all adhesives was calculated by determining the area under the bond slip curves derived from the FRP-to-concrete joint tests. Generally, the addition of GNP 0.075% to the control CBA mix increased its fracture energy by 178%. However, the epoxy resin used still had a fracture energy which was 359% higher than the GNP 0.075% mix.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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